

Physical Chemistry**Work Sheet - 08****Topic : Atomic Structure**

Fill in the blanks :

Species	No. of orbit (n)	radius (in Å) of orbit	Velocity of electron(m/sec)	Total energy of electron (eV)	Pot.Energy of electron (eV)	Kinetic Energy of electron (eV)
He ⁺	2					
Li ⁺²		1.587				
Be ⁺³			4.36×10^6			
H					-6.8	
He ⁺						3.4

Physical Chemistry**Work Sheet - 09****Topic : Atomic Structure**

Fill in the blanks

(R is Rydberg's constant)

Series of H-Spectrum	Region	Longest Wavelength of Series	Wavelength of Highest Energy Line	Series Limit (Wavelength)	Wavelength of n^{th} line
	UV				(3 rd line)
				4R	(5 th line)
Brackett series					(4 th line)
Pfund series					(2 nd line)
	Infrared			9/R	(2 nd line)